WORKED SOLUTION

STEP II 2009, Question 5

Pure • Nested Square Roots

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q5

The key idea

The opening expansion is the whole question in disguise: it tells us that $x+2\sqrt{x-1}$ is a perfect square, and the same trick shows $x-2\sqrt{x-1}$ is one too. Once the nested roots are unpicked, every integrand collapses to something elementary. The one thing to watch is that $\sqrt{a^2}=|a|$, not a, so we must keep track of signs, and that is exactly what makes part (ii) interesting.

Solution

The expansion

For $x \geq 1$,

$$(\sqrt{x-1}+1)^2 = (x-1)+2\sqrt{x-1}+1 = x+2\sqrt{x-1}.$$

In the same way $(\sqrt{x-1}-1)^2 = x-2\sqrt{x-1}$. Taking non-negative square roots, and remembering that $\sqrt{a^2}=|a|$,

$$\sqrt{x+2\sqrt{x-1}} = \sqrt{x-1}+1, \qquad \sqrt{x-2\sqrt{x-1}} = |\sqrt{x-1}-1|.$$

The first needs no modulus because $\sqrt{x-1}+1>0$. The second does: $\sqrt{x-1}-1$ is negative for $1 \leq x < 2$ and non-negative for $x \geq 2$.

Part (i)

On $[5,10]$ we have $\sqrt{x-1} \geq 2 > 1$, so $|\sqrt{x-1}-1| = \sqrt{x-1}-1$. The numerator is therefore

$$(\sqrt{x-1}+1)+(\sqrt{x-1}-1) = 2\sqrt{x-1},$$

and the integrand is simply 2. Hence

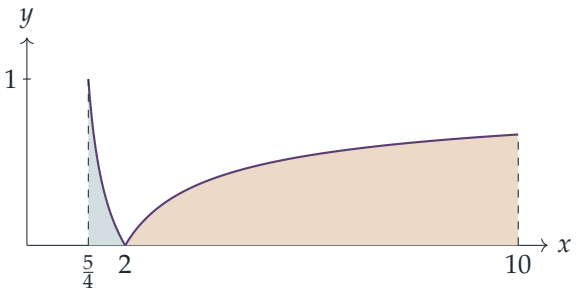
$$\int_5^{10} \frac{\sqrt{x+2\sqrt{x-1}}+\sqrt{x-2\sqrt{x-1}}}{\sqrt{x-1}}\,\mathrm{d}x = \int_5^{10} 2\,\mathrm{d}x = 10.$$

Part (ii)

By the expansion,

$$y = \frac{|\sqrt{x-1}-1|}{\sqrt{x-1}} = \left|1 - \frac{1}{\sqrt{x-1}}\right|.$$

So $y \geq 0$ throughout, and the curve touches the x-axis at $x = 2$, where $\sqrt{x-1} = 1$.



Ask yourself Why can't we just drop the square root and integrate $1 - \dfrac{1}{\sqrt{x-1}}$ over the whole range?

Because $\sqrt{x - 2\sqrt{x-1}}$ is never negative, but $\sqrt{x-1} - 1$ is negative for $x < 2$. Ignoring the modulus would count the piece on $[\frac{5}{4}, 2]$ as negative area. So we split at $x = 2$.

On $[\frac{5}{4}, 2]$, $\sqrt{x-1} \leq 1$, so $y = \dfrac{1}{\sqrt{x-1}} - 1$, and

$$\int_{5/4}^{2} \left(\frac{1}{\sqrt{x-1}} - 1\right) \mathrm{d}x = \left[2\sqrt{x-1} - x\right]_{5/4}^{2} = (2-2) - (1 - \tfrac{5}{4}) = \tfrac{1}{4}.$$

On $[2, 10]$, $\sqrt{x-1} \geq 1$, so $y = 1 - \dfrac{1}{\sqrt{x-1}}$, and

$$\int_{2}^{10} \left(1 - \frac{1}{\sqrt{x-1}}\right) \mathrm{d}x = \left[x - 2\sqrt{x-1}\right]_{2}^{10} = (10-6) - (2-2) = 4.$$

The total area is therefore $\frac{1}{4} + 4 = \frac{17}{4}$.

Part (iii)

The second root in the numerator has a different shape, but the same trick works with $x + 1$ in place of $x - 1$:

$$(\sqrt{x+1} - 1)^2 = (x+1) - 2\sqrt{x+1} + 1 = x - 2\sqrt{x+1} + 2.$$

For $x \geq \frac{5}{4}$ we have $\sqrt{x+1} > 1$, so $\sqrt{x - 2\sqrt{x+1} + 2} = \sqrt{x+1} - 1$, with no modulus needed. Writing $\sqrt{x^2-1} = \sqrt{x-1}\,\sqrt{x+1}$ (valid since both factors are positive for

$x > 1$), the integrand becomes

$$\frac{(\sqrt{x-1}+1)+(\sqrt{x+1}-1)}{\sqrt{x-1}\sqrt{x+1}} = \frac{\sqrt{x-1}+\sqrt{x+1}}{\sqrt{x-1}\sqrt{x+1}} = \frac{1}{\sqrt{x+1}} + \frac{1}{\sqrt{x-1}}.$$

The two ± 1 terms cancel, which is the point of the slightly odd-looking numerator. Hence

$$\begin{aligned}\int_{5/4}^{10}\Big(\frac{1}{\sqrt{x+1}} + \frac{1}{\sqrt{x-1}}\Big)\mathrm{d}x &= \Big[2\sqrt{x+1}+2\sqrt{x-1}\Big]_{5/4}^{10} \\ &= (2\sqrt{11}+6)-(3+1) \\ &= 2\sqrt{11}+2,\end{aligned}$$

using $\sqrt{9/4} = \frac{3}{2}$ and $\sqrt{1/4} = \frac{1}{2}$ at the lower limit.

Checks

- At $x = 5$: $\sqrt{5+4} = 3$ and $\sqrt{5-4} = 1$, so the integrand in (i) is $(3+1)/2 = 2$, as claimed.
- In (ii), at $x = \frac{5}{4}$ the curve has height $|1-2| = 1$ and at $x = 10$ height $1 - \frac{1}{3} = \frac{2}{3}$, matching the sketch; the first piece is a small sliver of area $\frac{1}{4}$, which is plausible.
- Simpson's rule on the original integrands gives 10.000, 4.250 and 8.6332, agreeing with 10, $\frac{17}{4}$ and $2\sqrt{11}+2 \approx 8.6332$.

Common slips

- Writing $\sqrt{x-2\sqrt{x-1}} = \sqrt{x-1}-1$ for all x. In (ii) this gives $4 - \frac{1}{4} = \frac{15}{4}$ instead of $\frac{17}{4}$.
- In (iii), not spotting that $x - 2\sqrt{x+1} + 2$ is also a perfect square, and trying a substitution instead.

Beyond STEP

The identity $\sqrt{a^2} = |a|$ is a small example of a big theme in analysis: functions defined by formulas can hide corners and changes of sign, and you must look at where an expression changes sign before integrating it. Denesting radicals such as $\sqrt{x+2\sqrt{x-1}}$ also has a surprisingly deep algebraic theory, connected to field extensions. For the analysis side, Abbott's *Understanding Analysis* is a good next step.

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